Chapter 1: Numerical Methods

We introduce a locally mass conserving numerical framework. We solve the static mechanics system with conforming mixed finite element method, which is adjusted to be locally mass conservative. We solve the transport system with finite volume scheme, which is a mass conservative scheme.

1.1 Quadrilateral Mesh

We consider a bounded computational domain $\Omega \subset \mathbb{R}^2$. Here, $\partial\Omega$ denotes the Lipschitz-continuous boundary of the domain. Let $\mathcal{T}_h$ be a conforming mesh of quadrilaterals covering Ω. The quadrilateral $E \in \mathcal{T}_h$ is referred as element in finite element context, while referred as cell in finite volume context. Each quadrilateral E is assumed to be strictly convex, not degenerate into a triangle, line or point. For ease of implementation, we use a logically rectangular mesh, where each element(cell) is indexed using Cartesian coordinates $\{i, j\}$. We show a general quadrilateral element(cell) in Figure 1.1. The edges of element are denoted by e_i, $i \in \{0, 1, 2, 3\}$ in the counterclockwise direction. The vertices are represented as $v_i = e_i \cap e_{i+1}$ in the sense of modulo. Here, $\boldsymbol{\nu}_i$ and $\boldsymbol{\tau}_i$ are used to represent the unit normal and the unit tangent vector respectively. The unit normal vector $\boldsymbol{\nu}_i$ points outerwards the quadrilateral. The tangent $\boldsymbol{\tau}_i$ is generated by rotating $\boldsymbol{\nu}_i$ counterclockwise by right-hand rule.

1.2 Mixed Formulation and Conforming Element

We use mixed finite element method (MFEM) for both Darcy and Stokes flow. We write mixed variational form for Darcy and Stokes flow separately and define conforming element with direct serendipity space.

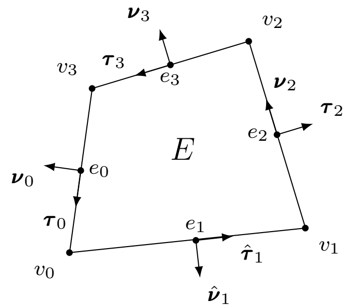

Figure 1.1: An exhibition of a general non-degenerate quadrilateral element(cell).

1.2.1 Stokes Problem

We begin with the strong form of the static Stokes problem as

$$
\begin{aligned}
-\nabla \cdot \boldsymbol{\tau}(\mathbf{u}) + \nabla p &= \mathbf{f}, \\
\nabla \cdot \mathbf{u} &= 0,
\end{aligned}
\tag{1.1}
$$

where $\boldsymbol{\tau}$ is the deviatoric stress similar to (**??**) as $\boldsymbol{\tau}(\mathbf{u}) = 2\big(\mathcal{D}\mathbf{u} - \frac{1}{3}\nabla \cdot \mathbf{u}\mathbf{I}\big)$. Let test function space

$$
V = \{\mathbf{v} \in (H^1(\Omega))^2 : \mathbf{v} = 0 \quad \text{on} \quad \Gamma_u\}
\tag{1.2}
$$

We derive weak form

$$
\begin{aligned}
(\nabla \mathbf{u}, \nabla \mathbf{v})_\Omega - (p, \nabla \cdot \mathbf{v})_\Omega = (\mathbf{f}, \mathbf{v})_\Omega &+ \langle \tau^T \nabla \mathbf{u} \nu, \mathbf{v} \cdot \tau \rangle_{\Gamma_i}, \\
&+ \langle \nu^T \nabla \mathbf{u} \nu - p_0, \mathbf{v} \cdot \nu \rangle_{\Gamma_i} - \langle \nabla \mathbf{u}_0, \nabla \mathbf{v} \rangle_{\Gamma_u} \\
(\nabla \cdot \mathbf{u}, \omega) = 0.&
\end{aligned}
\tag{1.3}
$$

Let stress $\mathbf{s} \cdot \tau = \tau^T \nabla \mathbf{u} \nu$ and $\mathbf{s} \cdot \nu = \nu^T \nabla \mathbf{u} \nu - p_0$, we have stress boundary condition term defined on Γ_i which is $< \mathbf{s}, \mathbf{v} >_{\Gamma_i}$. We employ a test space

$$
\begin{aligned}
(\nabla \mathbf{u}_h, \nabla \mathbf{v}_h)_{\Omega_h} - (p_h, \nabla \cdot \mathbf{v}_h)_{\Omega_h} &= (\mathbf{f}, \mathbf{v}_h)_{\Omega_h} + \langle \mathbf{s}, \mathbf{v}_h \rangle_{\Gamma_{i,h}} - \langle \nabla \mathbf{u}_{0,h}, \nabla \mathbf{v}_h \rangle_{\Gamma_{u,h}}, \\
(\nabla \cdot \mathbf{u}, \omega) &= 0.
\end{aligned}
\tag{1.4}
$$

We can separate normal and tangent part of stress on the boundary easily when physical boundary align with x-y axis. In the situation when a rectangular computational domain is used, we can mix stress and dirichlet boundary condition even further.

1.2.2 H^1 Conforming Element

1.2.3 Darcy Problem

$$\begin{aligned} \mathbf{u} + \nabla p &= \mathbf{g} \\ \nabla \cdot \mathbf{u} &= 0 \end{aligned} \tag{1.5}$$

weak form

$$\begin{aligned} (\mathbf{u}, \mathbf{v})_\Omega - (p, \nabla \cdot \mathbf{v})_\Omega &= (\mathbf{g}, \mathbf{v})_\Omega - \langle p_0, v_n \rangle_{\Gamma_i} - \langle \mathbf{u}_0, \mathbf{v} \rangle_{\Gamma_u} \\ (\nabla \cdot \mathbf{u}, \omega) &= 0 \end{aligned} \tag{1.6}$$

Test space $\{\mathbf{v} \in H(\mathrm{div}, \Omega) : v_n = 0 \text{ on } \Gamma_u\}$.

1.2.4 $H(\mathbf{div})$ Conforming Element

1.2.5 Coupled Degenerage System

scaled nondimensionalized weak form let $\check{\sigma} = \mathcal{D}\mathbf{v} - \frac{1}{3}\nabla \cdot \mathbf{v}\mathrm{I}$

$$\begin{aligned} (2\phi_s \check{\sigma}(\mathbf{v}_{s,h}), \nabla \Psi_s) - (\check{q}_h, \nabla \cdot \Psi_s) &= -(\phi_s \check{\mathbf{f}}, \Psi_s) \\ (\nabla \cdot \check{\mathbf{v}}_{s,h}, \omega) + \left(\frac{\hat{\phi}_f}{\phi_s}\check{q}_h, \omega\right) - \left(\frac{\hat{\phi}_f^{1/2}}{\phi_s}\check{q}_{f,h}, \omega\right) &= 0 \\ (\check{\mathbf{v}}_{r,h}, \Psi_r) - (\check{q}_{f,h}, \hat{\phi}_f^{-1/2}\nabla \cdot (\phi_f^{1+\Theta}\check{\mathbf{v}}_r)) &= 0 \\ (\hat{\phi}_f^{-1/2}\nabla \cdot (\phi_f^{1+\Theta}\check{\mathbf{v}}_r), \omega_f) + \left(\frac{1}{\phi_s}\check{q}_{f,h}, \omega_f\right) - \left(\frac{\hat{\phi}_f^{1/2}}{\phi_s}\check{q}_h, \omega_f\right) &= 0 \end{aligned} \tag{1.7}$$

1.3 Direct Serendipity Space

The classical serendipity space suffer from poor approximation on quadrilaterals. We define local shape functions directly on quadrilaterals according to Arbogast et al. (2022). H^1 and $H(\mathrm{div})$ conforming direct mixed finite elemenent space are constructed with the direct serendipity suplemental function space.

To begin with, we introduce some notations and distance auxilliary functions. Let $\mathbb{P}_r(\omega)$ denote the space of polynomials of degree up to r on $\omega \subset \mathbb{R}^d$. The dimension of $\mathbb{P}_r$ can be calculated as

$$\dim \mathbb{P}_r(\mathbb{R}^d) = \binom{r+2}{2} = \frac{(r+d)!}{r!d!}. \tag{1.8}$$

We define the linear polynomial $\lambda_i(\mathbf{x})$ giving the distance of $\mathbf{x} \in \mathbb{R}^2$ to edge e_i as

$$\lambda_i(\mathbf{x}) = -(\mathbf{x} - \mathbf{x}_i^*) \cdot \boldsymbol{\nu}_i, \quad i = 0, 1, 2, 3, \tag{1.9}$$

where $\mathbf{x}_i^* \in e_i$ is any point on the edge e_i. The value of distance function λ_i keeps positive as long as $\boldsymbol{x}$ is in the interior of the quadrilateral. We define two additional distance functions giving the distance of $\mathbf{x} \in \mathbb{R}^2$ to diagonal line d_i joining v_i with v_{i+2}, $i \in 0, 1$. Let $\nu_{d,i}$ be the unit normal vector to the diagonal line d_i. The direction of the $\nu_{d,i}$ is defined by counterclockwise rotating of the vector pointing from v_i to v_{i+2}. The resulting diagonal distance function is defined as

$$\lambda_{d,i}(\mathbf{x}) = -(\mathbf{x} - \mathbf{x}_{d,i}^*) \cdot \boldsymbol{\nu}_{d,i}, \quad i = 1, 2, \tag{1.10}$$

where $\mathbf{x}_{d,i}^*$ is any point on the diagonal line d_i. For simplicity, we use corner v_i in later computation.

Now we present set of nodal basis functions in first and second order direct serendipity space $\mathcal{DS}_1$ and $\mathcal{DS}_2$ defined by Arbogast et al. (2022).

1.3.1 Direct Serendipity Space $\mathcal{DS}_2$

We supplement polynomial space $\mathbb{P}_r(E), r \geq 2$ with two linearly independent functions, $\phi_{s,1}(\mathbf{x})$ and $\phi_{s,2}(\mathbf{x})$. We denote the supplemental function space as $\mathbb{S}_r^{\mathcal{DS}}(E) = \text{span}\{\phi_{s,1}, \phi_{s,2}\}$. We present the second order direct serendipity space as

$$\mathcal{DS}_r(E) = \mathbb{P}_r(E) \oplus \mathbb{S}_r^{\mathcal{DS}}(E). \tag{1.11}$$

The supplemental functions are defined as

$$\phi_{s,i} = \lambda_{i+1}\lambda_{i+2}R_{i,i+2}, \quad i \in \{1, 2\}, \tag{1.12}$$

where i is in the sense of modulo. We use a simple definition for rational function $R_{i,i+2}$ as

$$R_{i,i+2} = \frac{\lambda_i - \lambda_{i+2}}{\lambda_i + \lambda_{i+2}}, \tag{1.13}$$

which satisfies the conditions as

$$R_{i,i+2}|_{e_i} = -1, \quad R_{i,i+2}|_{e_{i+2}} = 1. \tag{1.14}$$

We continue to define

$$R_i = \frac{1}{2}\Big(1 - (-1)^{\lfloor i/2 \rfloor} R_{i,i+2}\Big) = \frac{\lambda_{i+2}}{\lambda_i + \lambda_{i+2}}, \tag{1.15}$$

where i is understood $\pmod 2$. Here, $R_i|_{e_i} = 1$, and $R_i|_{e_{i+2}} = 0$ by definition. In Figure 1.2, we draw R_0 on a general quadrilateral, which evaluates 1 on the edge e_0, and evaluates 0 on the opposite edge e_2.

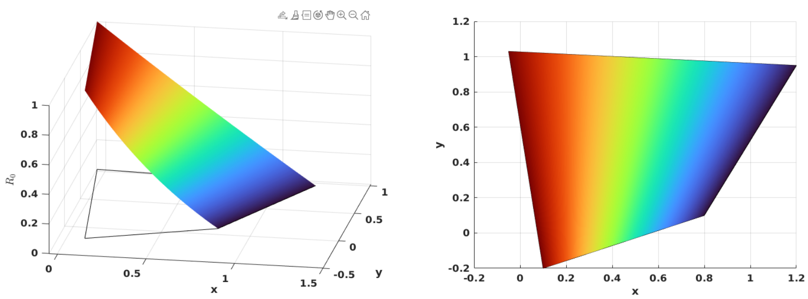

Figure 1.2: Rational function R_0.

The edge nodal basis functions for $\mathcal{DS}_2$ are constructed as

$$\varphi_{e,i} = \frac{\lambda_{i+1}\lambda_{i+3}R_i}{\lambda_{i+1}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})R_i(\mathbf{x}_{e,i})}, \tag{1.16}$$

where $\mathbf{x}_{e,i}$ is the middle point of the edge e_i, the i index is understood in the sense of modulo of 4. In Figure 1.3, we present $\varphi_{e,0}$ on a general quadrilateral.

The vertex nodal basis functions for $\mathcal{DS}_2$ are constructed as

$$\varphi_{v,i} = \frac{\lambda_{i+2}\lambda_{i+3} - \sum_{k=i}^{i+1} \lambda_{i+2}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})\varphi_{e,i}}{\lambda_{i+2}(\mathbf{x}_{v,i})\lambda_{i+3}(\mathbf{x}_{v,i}) - \sum_{k=i}^{i+1} \lambda_{i+2}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})\varphi_{e,i}(\mathbf{x}_{v,i})}, \tag{1.17}$$

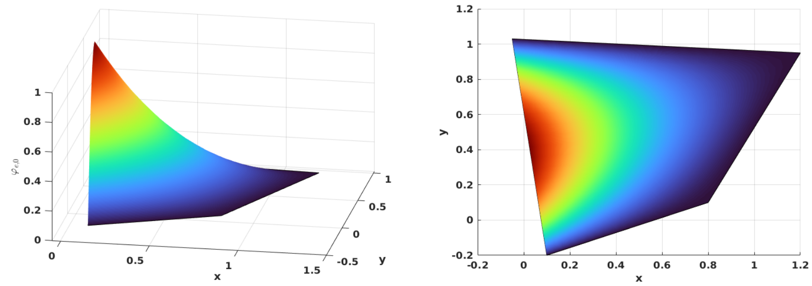

Figure 1.3: Edge nodal basis function $\varphi_{e,0}$ in $\mathcal{DS}_2$.

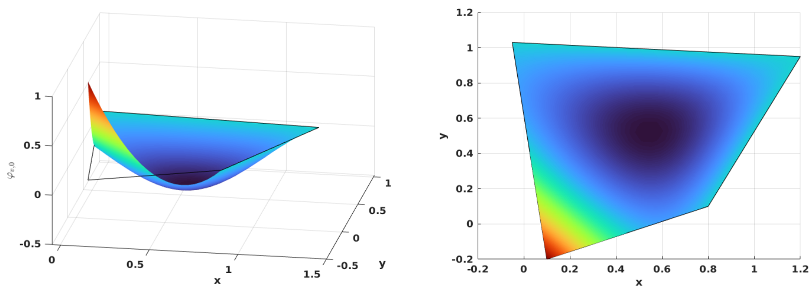

Figure 1.4: Vertex nodal basis function $\varphi_{v,0}$ in $\mathcal{DS}_2$.

where $\mathbf{x}_{v,i}$ is the coordinates of the vertex node v_i. The i index is understood in the sence of module of 4 as well. In Figure 1.4, we show $\varphi_{v,0}$ on a general quadrilateral.

We conclude that unisolvence follows directly from the constructed nodal basis functions.

1.3.2 Direct Serendipity Space $\mathcal{DS}_1$

The supplemental space for $\mathcal{DS}_1$ differs from that for $\mathcal{DS}_2$. We present the first order serendipity as

$$\mathcal{DS}_1(E) = \mathbb{P}_1(E) \oplus \operatorname{span}\{R\}, \tag{1.18}$$

where $R(\mathbf{x}_{v,0}) = R(\mathbf{x}_{v,2}) = -R(\mathbf{x}_{v,1}) = -R(\mathbf{x}_{v,3}) = 1$. The construction of rational funcion R is not unique. We present one way to construct $R(\mathbf{x})$ with edge nodal basis function (1.16) in $\mathcal{DS}_2$ as

$$R(\mathbf{x}) = r(\mathbf{x}) - \sum_{i=0}^{3} r(\mathbf{x}_{e,i})\varphi_{e,i}(\mathbf{x}), \tag{1.19}$$

where auxiliary function $r(\mathbf{x})$ is defined as

$$r(\mathbf{x}) = \frac{\lambda_2(\mathbf{x})\lambda_3(\mathbf{x})}{\lambda_2(\mathbf{x}_{v,0})\lambda_3(\mathbf{x}_{v,0})} - \frac{\lambda_0(\mathbf{x})\lambda_3(\mathbf{x})}{\lambda_0(\mathbf{x}_{v,1})\lambda_3(\mathbf{x}_{v,1})} + \frac{\lambda_0(\mathbf{x})\lambda_1(\mathbf{x})}{\lambda_0(\mathbf{x}_{v,2})\lambda_1(\mathbf{x}_{v,2})} - \frac{\lambda_1(\mathbf{x})\lambda_2(\mathbf{x})}{\lambda_1(\mathbf{x}_{v,3})\lambda_2(\mathbf{x}_{v,3})}. \tag{1.20}$$

In Figure 1.5, we plot the rational function on a general quadrilateral.

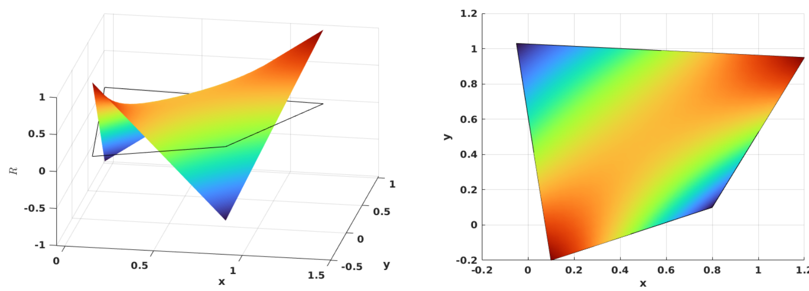

Figure 1.5: Rational function in $\mathcal{DS}_1$.

We define vertex nodal basis function accordingly as

$$\varphi_{v,i}(\mathbf{x}) = \frac{\lambda_{d,m}(\mathbf{x}) - \frac{1}{2}\lambda_{d,m}(\mathbf{x}_{v,i+2})(1 + (-1)^i R(\mathbf{x}))}{\lambda_{d,m}(\mathbf{x}_{v,i}) - \lambda_{d,m}(\mathbf{x}_{v,i+2})}, \tag{1.21}$$

where $m \equiv i + 1 \pmod 2$, and index i is understood in the sence of modulo of 4. In Figure 1.6, we show the vertex nodal basis function $\varphi_{v,0}$ in $\mathcal{DS}_1$ on a general quadrilateral, which evaluates 1 at vertex v_0 and 0 at other vertex.

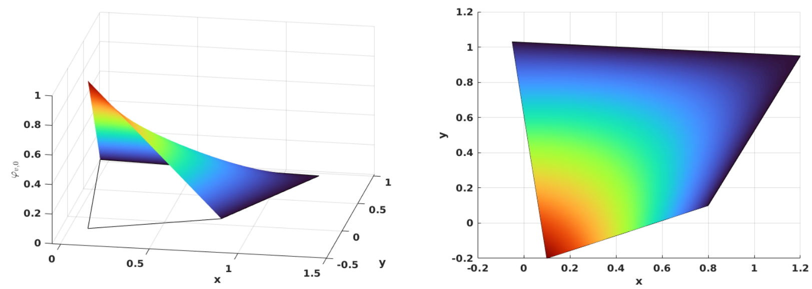

Figure 1.6: Vertex nodal basis function in $\mathcal{DS}_1$.

1.4 Linear System

1.4.1 Saddle Point Problem

1.4.2 Degenerate Coupled Saddle Point Problem

The linear system corresponding to (insert eqcite) has the form

$$\begin{bmatrix} \mathbf{A}_s & -\mathbf{B}_s & \mathbf{0} & \mathbf{0} \\ \mathbf{B}_s^T & \mathbf{C}_s & \mathbf{0} & \mathbf{K} \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_d & -\mathbf{B}_d \\ \mathbf{0} & \mathbf{K} & \mathbf{B}_d^T & \mathbf{C}_d \end{bmatrix} \begin{bmatrix} \mathbf{x}_s \\ \mathbf{y}_s \\ \mathbf{x}_d \\ \mathbf{y}_d \end{bmatrix} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{g}_s \\ \mathbf{f}_d \\ \mathbf{g}_d \end{bmatrix}, \tag{1.22}$$

wherein the solution represents the degrees of freem of primary variables with respect to the bases of the finite element spaces. (Insert details of calculation of each blocked matrix). We form a double saddle point system by symmetrically permutate original linear system,

$$\begin{bmatrix} \mathbf{A}_s & \mathbf{0} & -\mathbf{B}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_d & \mathbf{0} & -\mathbf{B}_d \\ \mathbf{B}_s^T & \mathbf{0} & \mathbf{C}_s & \mathbf{K} \\ \mathbf{0} & \mathbf{B}_d^T & \mathbf{K} & \mathbf{C}_d \end{bmatrix} \begin{bmatrix} \mathbf{x}_s \\ \mathbf{x}_d \\ \mathbf{y}_s \\ \mathbf{y}_d \end{bmatrix} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{f}_d \\ \mathbf{g}_s \\ \mathbf{g}_d \end{bmatrix}. \tag{1.23}$$

We group these block matrix to form a concise version,

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_d \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} -\mathbf{B}_s & \mathbf{0} \\ \mathbf{0} & -\mathbf{B}_d \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} -\mathbf{C}_s & -\mathbf{K} \\ -\mathbf{K} & -\mathbf{C}_d \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{f}_d \end{bmatrix}, \quad \mathbf{G} = \begin{bmatrix} -\mathbf{g}_s \\ -\mathbf{g}_d \end{bmatrix}, \tag{1.24}$$

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^T & \mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{F} \\ \mathbf{G} \end{bmatrix}, \tag{1.25}$$

where $\mathbf{A}$ is a positive definite matrix. However, $\mathbf{B}$ is usually a singular matrix when no melting region presents. We will present a modified Uzawa iteration algorithm to deal with this coupled degenerate singular system. Inexact Uzawa algorithm has been introduced in the previous literature Elman and Golub (1994) and Bramble et al. (1997),

Algorithm 1 Preconditioned Inexact Uzawa Iteration

1: Initialize $\mathbf{x}_0 = \mathbf{0}$ and $\mathbf{y}_0 = \mathbf{0}$
2: Update $\mathbf{x}_{i+1} = \mathbf{x}_i + \mathbf{A}^{-1}(\mathbf{F} - (\mathbf{A}\mathbf{x}_i + \mathbf{B}\mathbf{y}_i))$
3: Obtain $\mathbf{r} = \mathbf{B}^T\mathbf{x}_{i+1} + \mathbf{C}\mathbf{y}_i - \mathbf{G}$
4: Obtain $\tilde{\mathbf{y}} = \text{argmin}_{\mathbf{x}\in V}||(\mathbf{B}^T\mathbf{A}^{-1}\mathbf{B} - \mathbf{C})^{-1}\tilde{\mathbf{y}} - \mathbf{r}||$
5: Update $\mathbf{y}_{i+1} = \mathbf{y}_i + \tilde{\mathbf{y}}$

Works Cited

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